

Use the vocabulary in the table below to complete each statement.

Line Plots	Histograms	Stem-and-leaf plots	Box plots	Bar Graph	Circle Graph
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1. A bar graph is a visual display of categorical data values where each category is represented by a bar whose height represents the number in the category.
2. Histograms are best for larger sets of data, data over a large range, or continuous data. One disadvantage is that the display cannot be used to calculate measures of center and spread since individual values cannot be determined from the display.
3. A circle graph is a visual display of categorical data. The whole set of data is represented by the circle and its interior.
4. Box plots are helpful in visualizing the shape of a distribution and quickly identifying the five-number summary. This display cannot be used to find the mean.
5. Line plots are best for smaller sets of data and when the goal is to see each value over a scale, making it easier to calculate measures of center and spread.
6. Stem-and-leaf plots are helpful in visualizing the shape of a distribution, while still being able to see the individual values.

Jamal wants to create a graphical display of the shoe size of all 1457 students in his high school.

7. Which representation would be the best display for the data? Explain your reasoning.

Box plot or histogram because large sets of numerical data are best displayed in a box plot or histogram

8. Which other display should not be considered based on the size of the data?

Line plot

Penelope wanted to create a graphical display of the number of seeds in the seed packets she got to plant for her garden. Penelope got 25 seed packets comprised of tomatoes, lettuce, carrots, cucumbers, and squash packets. Each seed packet has at least 10 seeds in each packet.

9. Which representation would be the best display for the data? Explain your reasoning.

Box plot, line plot, steam-and-leaf

10. Are there any other display types that should not be considered based on the size of the data?

Histogram – small range of values